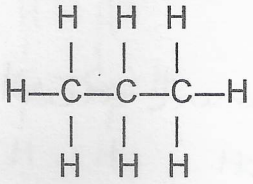
Naming Organic molecules Worksheet

1. Name the following and give the condensed structural formula:

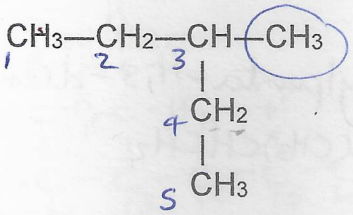
1.1



propane

$CH_3CH_2CH_3$

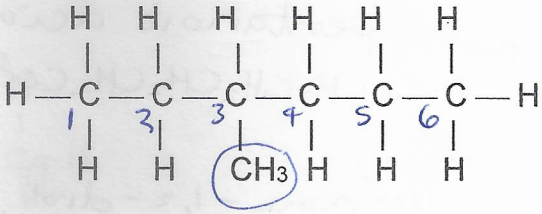
1.2



3-methylpentane

$CH_3CH_2CH(CH_3)CH_2CH_3$

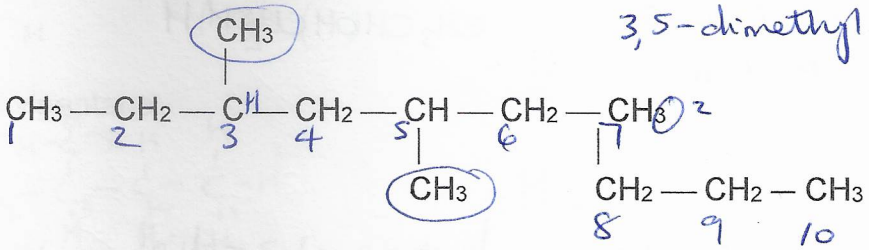
1.3



3-methylhexane

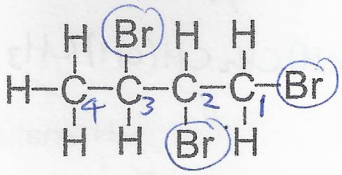
$CH_3CH_2CH(CH_3)CH_2CH_2CH_3$

1.4



3,5-dimethyldecane

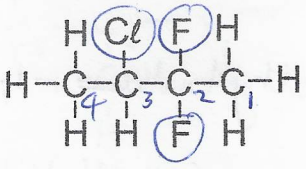
1.5



1,2,3-tribromobutane

$CH_3CHBrCHBrCH_2Br$

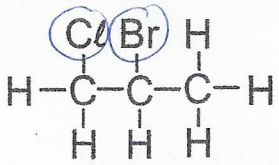
1.6



3-chloro-2,2-difluorobutane

$CH_3CHClCF_2CH_3$

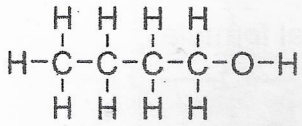
1.7



2-brano-1-chloropropane

$CH_2ClCHBrCH_3$

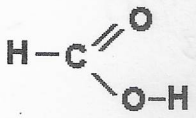
1.8



butan-1-ol

$CH_3CH_2CH_2CH_2OH$

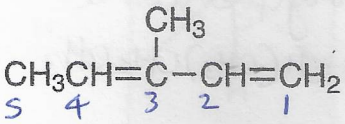
1.9



methanoic acid

$HCOOH$

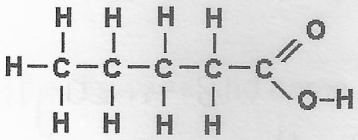
1.10



3-methylpenta-1,3-diene

$CH_3CHC(CH_3)CHCH_2$

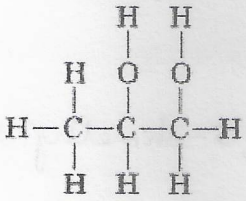
1.11



pentanoic acid

$CH_3CH_2CH_2CH_2COOH$

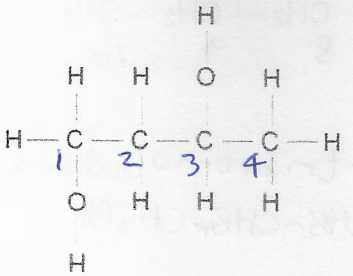
1.12



propane-1,2-diol

$CH_3CH(OH)CH_2(OH)$

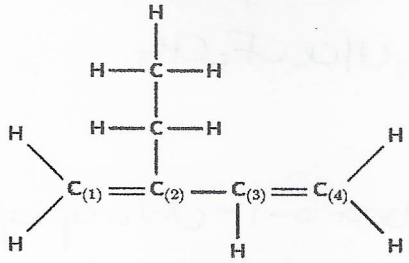
1.13



butane-1,3-diol

$CH_2(OH)CH_2CH(OH)CH_3$

1.14



2-ethylbutane-1,3-diene

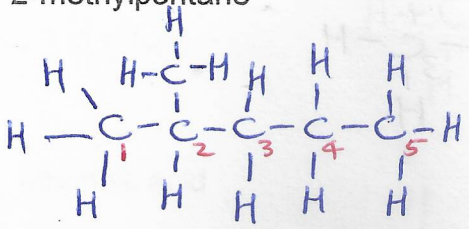
$CH_2C(CH_2CH_3)CHCH_2$

1.15 $CH_3CH_2CH_2COOH$

butanoic acid

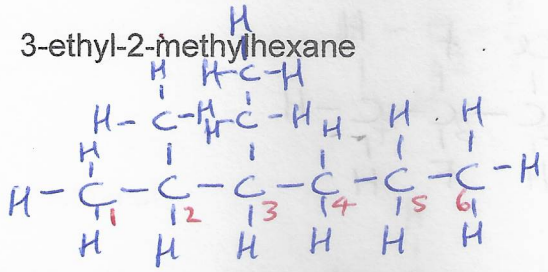
2. Draw the structural formula and molecular formula for the following.

2.1 2-methylpentane



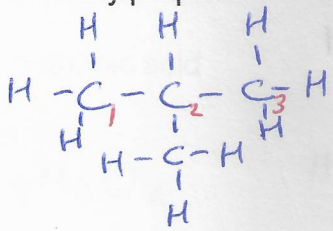
C_6H_{14}

2.2 3-ethyl-2-methylhexane



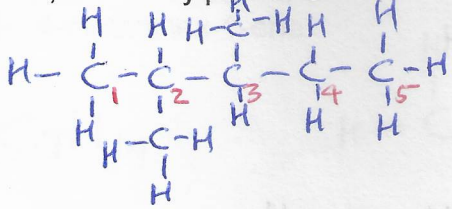
C_9H_{20}

2.3 2-methylpropane



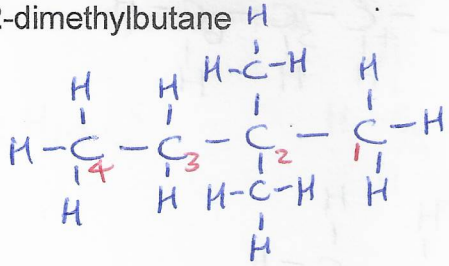
C_4H_{10}

2.4 2,3-dimethylpentane



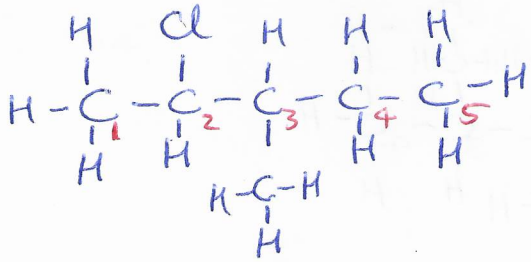
C_7H_{16}

2.5 2,2-dimethylbutane



C_6H_{14}

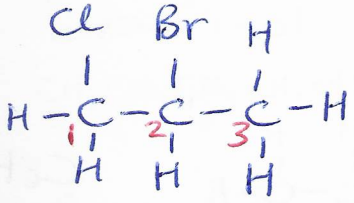
2.6 2-chloro-3-methylpentane



$C_6H_{13}Cl$

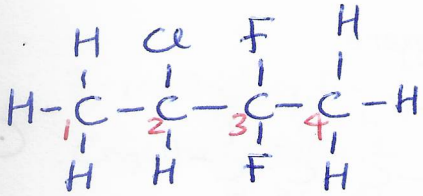
2.7 2-bromo-1-chloropropane

C_3H_6BrCl



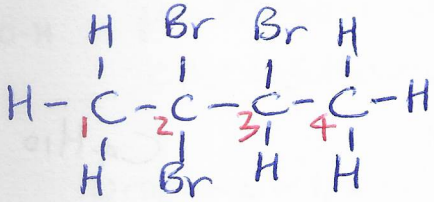
2.8 2-chloro-3,3-difluorobutane

$C_4H_7ClF_2$



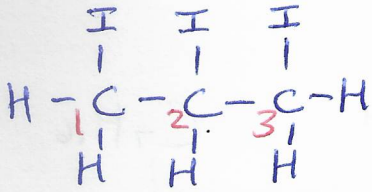
2.9 2,2,3-tribromobutane

$C_4H_7Br_3$



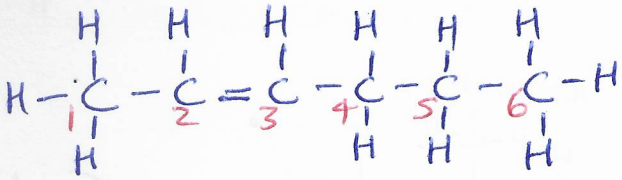
2.10 1,2,3-triiodopropane

$C_3H_5I_3$



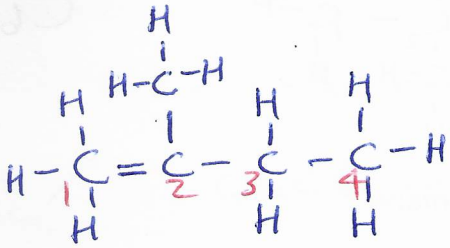
2.11 hex-2-ene

C_6H_{12}



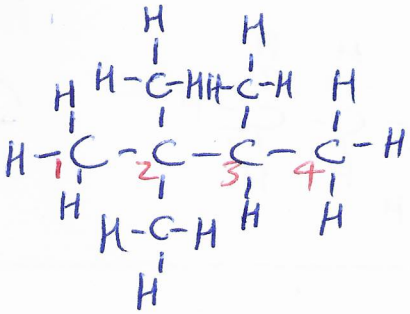
2.12 2-methylbut-1-ene

C_5H_{11}



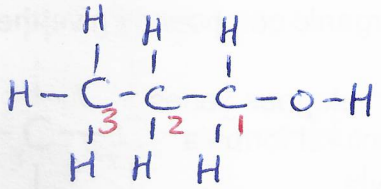
2.13 2,2,3-trimethylbutane

C_7H_{16}



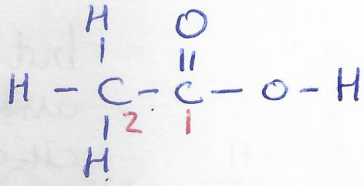
2.14 propan-1-ol

C_3H_8O



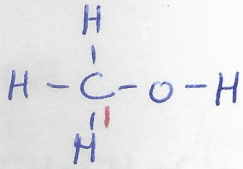
2.15 ethanoic acid

$C_2H_4O_2$



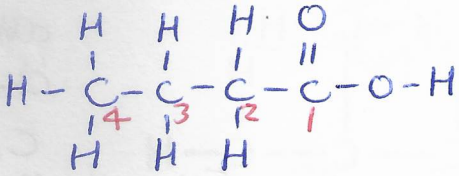
2.16 methanol

CH_4O



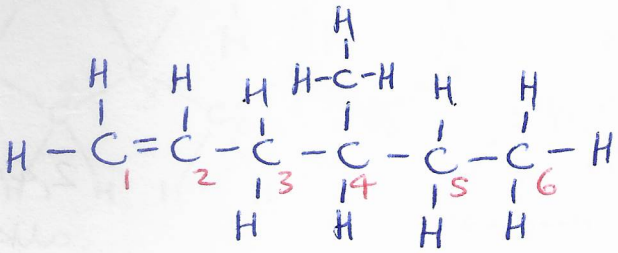
2.17 butanoic acid

$C_4H_8O_2$



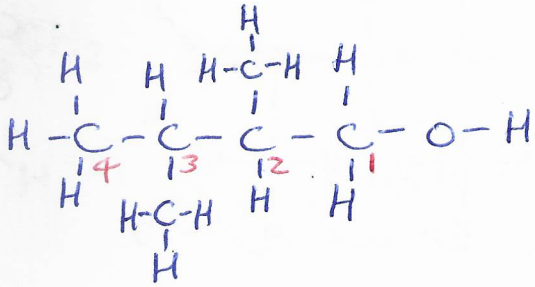
2.18 4-methylhex-1-ene

C_7H_{14}



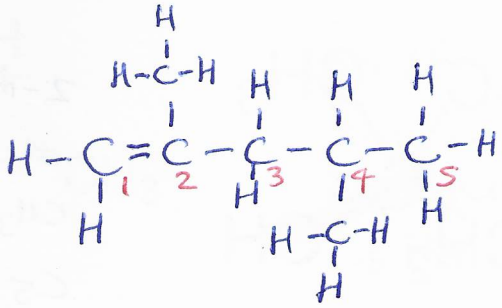
2.19 2,3-dimethylbutan-1-ol

$C_6H_{14}O$



2.20 2,4-dimethylpent-1-ene

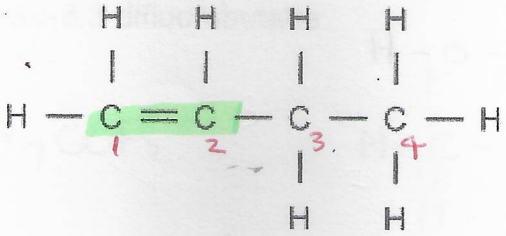
C_7H_{14}



3. For each of the following organic compounds give the:

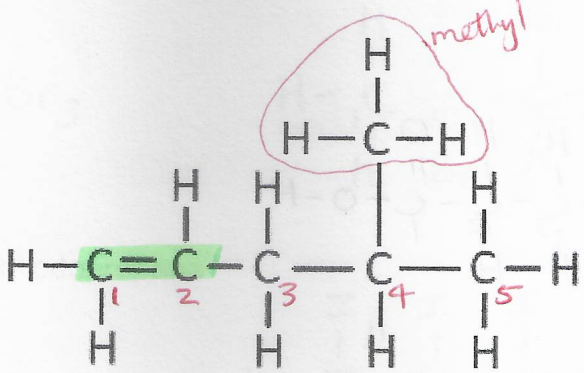
- IUPAC name
- The name of the homologous series
- The condensed structural formula
- The molecular formula
- Highlight the functional groups.

3.1



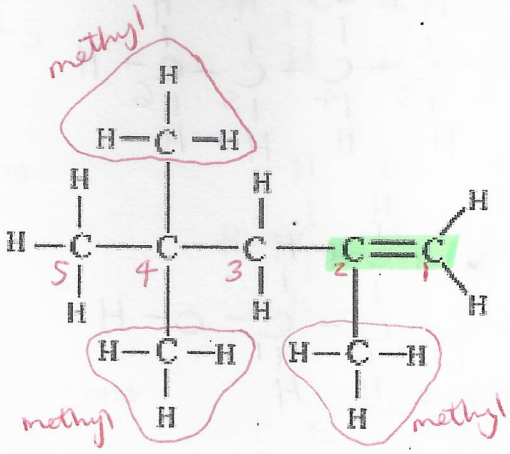
but-1-ene
alkenes
$CH_2CHCH_2CH_3$
C_4H_8

3.2



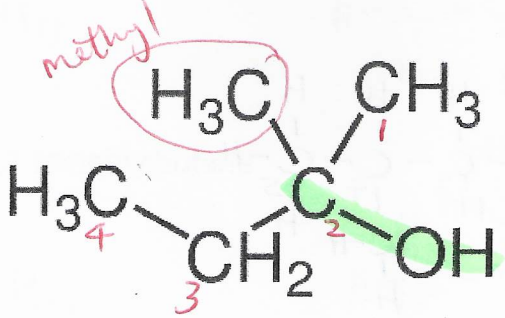
4-methylpent-1-ene
alkenes
$CH_2CHCH_2CH(CH_3)CH_3$
C_6H_{12}

3.3



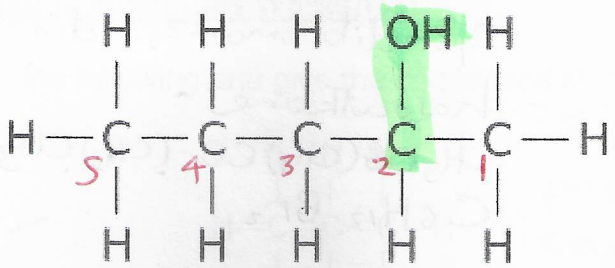
2,4,4-trimethylpent-1-ene
alkenes
$CH_2C(CH_3)CH_2C(CH_3)_2CH_3$
C_8H_{16}

3.4



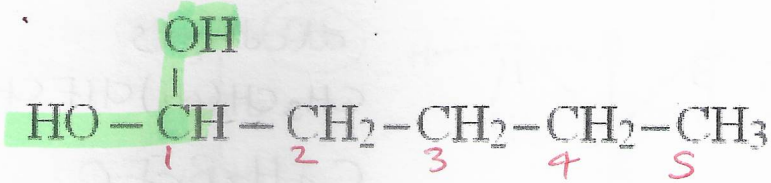
2-methylbutan-2-ol
alcohols
$CH_3COH(CH_3)CH_2CH_3$
$C_5H_{12}O$

3.5



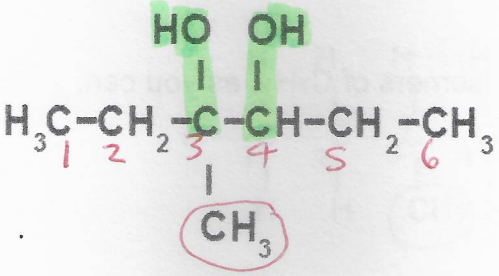
pentan-2-ol
alcohols
$CH_3CH(OH)CH_2CH_2CH_3$
$C_5H_{12}O$

3.6



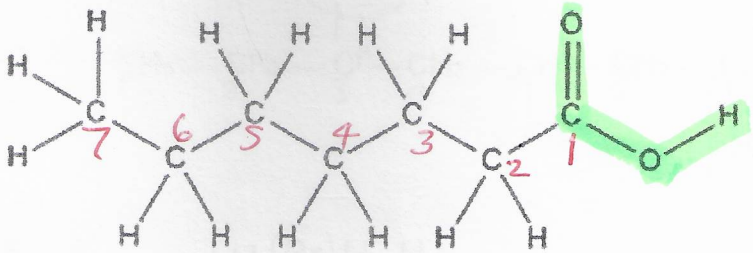
pentane-1,1-diol
alcohols
$CH(OH)_2CH_2CH_2CH_2CH_3$
$C_5H_{12}O_2$

3.7



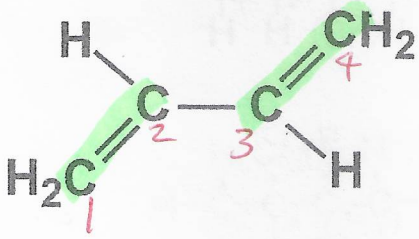
3-methylhexane-3,4-diol
alcohols
$CH_3CH_2COH(CH_3)C(OH)HCH_2CH_3$
$C_7H_{16}O_2$

3.8



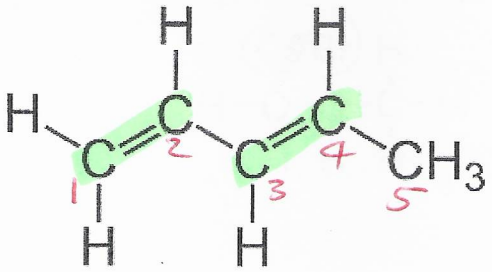
heptanoic acid
carboxylic acids
$CH_3CH_2CH_2CH_2CH_2CH_2COOH$
$C_7H_{14}O_2$

3.9



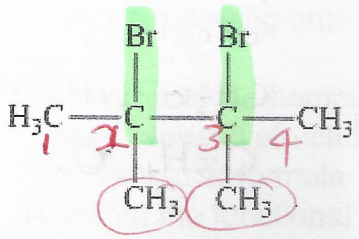
buta-1,3-diene
alkenes
$CH_2CHCHCH_2$
C_4H_6

3.10



penta-1,3-diene
alkenes
$CH_2CHCHCHCH_3$
C_5H_8

3.11



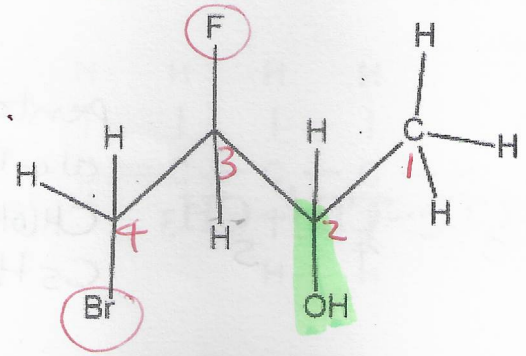
2,3-dibromo-2,3-dimethylbutane

haloalkanes

$CH_3CBr(CH_3)CBr(CH_3)CH_3$

$C_6H_{12}Br_2$

3.12



4-bromo-3-fluorobutan-2-ol

alcohols

$CH_3CH(OH)CHFCH_2Br$

C_4H_8BrFO

4. Draw the structure and name as many isomers of C_7H_{16} as you can.